

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: CRISISLTLSUM | Context: Ecuador — Humanitarian Crisis Social Media Analysts

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output ontology consists of (a) a binary relevance label conditioned on three conjunctive criteria — relevant, informative (facts vs. 'personal points of view'), and not repetitive [\[Q58\]](#) — and (b) a four-axis summary rubric (Coherence, Accuracy, Coverage, Overall) scored 1–5 [\[Q106, Q151–Q160\]](#). The Coherence rubric explicitly references English grammar and English-language understandability [\[Q153\]](#). For humanitarian coordinators, relevance criteria differ structurally: road/bridge access status to affected communities, informal community broadcasts, displacement movement indicators, alert level changes, and 'damnificados' counts at parroquia/cantón level are operationally salient (region YAML; elicitation A3), and arguably some 'personal points of view' from affected communities are highly informative for needs assessment — directly conflicting with the benchmark's exclusion of personal views [\[Q58\]](#). The four-axis summary rubric is partially transferable in abstract structure but its operational definitions are anchored to English [\[Q153\]](#) and contain no notion of humanitarian reporting conventions (UN OCHA SITREP format, numbered findings, agency acronym usage).

ID	Evidence
Q58	"A tweet is labeled as part of the timeline only if it satisfies all the following three conditions: relevant: talks about the same event indicated in the seed tweet, informative: provides facts about the event but not only contains personal points of view, not repetitive: brings in new information." (p.5)
Q106	"Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)
Q151	"During this task, you will be reading a timeline that consists of a sequence of tweets describing an event and eight different summaries for the event in the timeline. You will rate the quality of each of the eight summaries by four axes: coherence, accuracy, coverage, and overall quality." (p.21)
Q152	"The rubrics below give specific guidance on how each axis should be rated. Please read the rubrics carefully before continuing to the task." (p.21)
Q153	"For each summary, answer the question 'how coherent is the summary on its own?' on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors." (p.21)
Q154	"Score of 1: The summary is impossible to understand. Score of 2: The summary has many mistakes or confusing phrasing. Score of 3: The summary has some mistakes or confusing phrasing that made it hard to understand. Score of 4: The summary is mostly clear with only a few minor mistakes or confusing phrasing. Score of 5: The summary is perfectly clear." (p.21)
Q155	"For each summary, answer the question 'Does the factual information in the summary accurately match the information in the timeline?' on a scale of 1 to 5. A summary is accurate if it doesn't say things that aren't in the timeline, it doesn't mix up information, and generally is not misleading." (p.21)
Q156	"Score of 1: The summary is completely wrong, made up, or exactly contradicts what is written in the timeline. Score of 2: The summary says many things not mentioned in or contradicting the timeline. Score of 3: The summary says at least one crucial piece of information that is not in the timeline, or it contradicts something in the timeline. Score of 4: The summary says anything at all that is not mentioned in the timeline or contradicts something in the timeline. Score of 5: The summary has no incorrect statements or misleading implications." (p.21)
Q157	"For each summary, answer the question 'How well does the summary cover the important information in the timeline?' on a scale of 1 to 5. A good summary has good coverage if it mentions the main information from the timeline that's important to understand the event. A summary has poor coverage if someone reading only the summary would be missing several important pieces of information about the event in the timeline." (p.21)

Q158	"Score of 1: The summary contains no information relevant to the timeline. Score of 2: The summary is missing many important pieces of information required to understand the timeline. Score of 3: The summary is missing at least one crucial piece of information required to understand the timeline. Score of 4: The summary is missing some important information from the timeline. Score of 5: The summary covers all of the important information required to understand the timeline." (p.21)
Q159	"For each summary, answer the question "How good is the summary overall at representing the timeline?" on a scale of 1 to 5. This encompasses all of the above axes, as well as the information included in the summary and if it has helped you understand the event. If it's hard to find ways to make the summary better, give the summary a high score. If there are lots of different ways the summary could be better, give the summary a low score." (p.21)
Q160	"Score of 1: The summary is terrible. Score of 2: The summary is a pretty bad representation of the timeline and needs significant improvement. Score of 3: The summary is an okay representation of the timeline, but could be significantly improved. Score of 4: The summary is a pretty good representation of the timeline, but it's not perfect. Score of 5: The summary is an excellent representation of the timeline." (p.21)

Strengths

- The four-axis rubric (Coherence, Accuracy, Coverage, Overall) is a generic and well-structured summarization evaluation framework [Q106, Q151] that could plausibly be adapted to Spanish reference summaries with rewritten anchors; the Accuracy axis [Q155] aligns with humanitarian concerns about avoiding hallucinated content.

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Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: The binary 'part of timeline / not' label is too coarse for humanitarian use; the rubric axes are reusable in form but their English-anchored definitions [Q153] need redesign.

OO-2: Missing categories include: humanitarian-relevance sub-types (logistics access, needs signals, coordination signals, displacement), source-type tagging (official agency vs. community vs. media), and alert-escalation states (critical for IGEPN/SNGR volcanic monitoring) — none are represented [Q58, Q151].

OO-3: The 'informative = facts not personal points of view' criterion [Q58] encodes a US-journalism stance that excludes precisely the affected-community first-person testimony often most operationally salient in humanitarian monitoring.

OO-4: Stakeholder-driven taxonomy redesign is required, involving Cruz Roja Ecuatoriana and UN OCHA Ecuador analysts, to define a multi-label relevance schema and SITREP-aligned summary criteria.

OO-5: Structural and content validity are violated: the output space does not represent the constructs ('humanitarian operational relevance', 'situation report quality') the deployment cares about.

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WEB-21	HumVI confirms humanitarian NLP requires bespoke relevance taxonomies not covered by general crisis benchmarks. (https://arxiv.org/html/2410.06370)

Information Gaps

- Whether Cruz Roja/UN OCHA have published formal relevance schemas adaptable as a target ontology.

Requires Expert Verification

- Operational definition of 'important information' in Ecuadorian humanitarian SITREP conventions vs. benchmark's 'Coverage' axis.

Remediation

Gap 1: Binary 'part of timeline' criterion and four-axis rubric are anchored to English/US-journalism norms; humanitarian relevance categories (logistics access, community needs, displacement signals, alert escalation) are unrepresented.

Recommendation: Redesign the relevance schema as multi-label humanitarian-operational tagging (logistics, needs, displacement, coordination, casualties, infrastructure, alert-state) and rewrite the summary rubric to include SITREP genre conformance, bilingual coherence, and source-type provenance.